REGULATORY FRAMEWORK

Financial AI Regulatory Compliance Framework

Korea AI Basic Act, EU AI Act, FSS AI RMF Compliance Architecture Design Document

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Executive Summary — This document systematically maps the domestic and international regulatory requirements that the financial AI recommendation system (PLE-based) must comply with, and defines system-level response architectures for each requirement. It covers the Korea AI Basic Act (effective 2026.1.22), the Financial Services Commission (FSC) Integrated AI Guidelines 7 Principles, the Financial Supervisory Service (FSS) AI RMF, EU AI Act (Art. 13/14/15), and GDPR Art. 22, including end-to-end compliance architecture spanning audit trails, fairness monitoring, drift surveillance, herding detection, kill switches, opt-out, Human-in-the-Loop, and automated governance report generation.

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Financial AI Regulatory Environment Overview

Korea AI Basic Act

The **"Framework Act on the Development of Artificial Intelligence and Establishment of Trust"** (Act No. 20676), passed by the National Assembly in December 2024 and promulgated on January 21, 2025, took effect on **January 22, 2026**. It is the first comprehensive AI legislation in the Asia-Pacific region and adopts a Risk-Based Approach.

Core Legal Structure

Transparency Obligations (Art. 31 / Enforcement Decree Art. 22)
Prior notification of AI use, labeling of AI-generated content
Fine up to 30 million KRW for violations

Safety Assurance (Art. 32 / Enforcement Decree Art. 23)
Targets ultra-large models with training compute >= 10^{26} FLOPs
(Recommendation models are exempt — high-impact AI classification based on **usage domain** applies)

High-Impact AI Verification (Art. 33 / Enforcement Decree Art. 24)
Self-review obligation for operators
May request verification from MSIT (response within 30 days)

High-Impact AI Operator Duties (Art. 34 / Enforcement Decree Art. 26)
Establish risk management measures, user protection (explainability),
Public disclosure on website, etc.

High-Impact AI and Finance

The AI Basic Act defines high-impact AI as **"artificial intelligence that significantly affects or may pose risks to human life, physical safety, or fundamental rights"** (Art. 2, Item 4). The Enforcement Decree enumerates 11 domains, explicitly including the **finance (credit/lending)** sector.

Domain	Application Examples	Finance Relevance
Healthcare	AI used in medical devices and digital medical devices	Indirect
Finance (Credit/Lending)	**AI for credit scoring, loan assessment, and personal rights evaluation**	**Direct**
Employment	AI for hiring and HR decisions	None
Public Services	Eligibility verification, administrative decision-making AI	Indirect
Nuclear/Biometric	Nuclear facilities, biometric analysis AI	Indirect

Implications for Finance

AI used in **loan assessment and credit scoring** explicitly qualifies as high-impact AI under the law. Even if a financial AI recommendation system does not directly perform lending operations, AI applications in **product recommendation, suitability assessment, and customer classification** may be designated as high-impact AI through future Enforcement Decree amendments or FSC guidelines. Therefore, proactively building a governance framework **at the level of high-impact AI** is a rational strategy.

FSC Integrated AI Guidelines — 7 Principles

The Financial Services Commission published the **"Revised Financial AI Guidelines"** in alignment with the AI Basic Act. Scheduled for **Q1 2026 implementation** as self-regulatory standards per financial sector, they are organized around 7 principles.

1. Governance

Role and responsibility sharing among executives including CEO
AI risk management organization **independently separated** from planning/development

2. Legality

Compliance with AI Basic Act, Personal Information Protection Act, Credit Information Act

3. Auxiliary Nature

AI is a decision-support tool,
Financial institution bears final responsibility even for external models

4. Reliability

Model performance and bias management, explainability (XAI) assurance

5. Financial Stability

Third-party dependency, market herding risk assessment
Report to financial authorities on incidents

6. Good Faith

All customer-facing AI services require prior notification (broader than AI Basic Act)
Conflict of interest prevention, fairness criteria establishment and evaluation

7. Security

Data poisoning, model poisoning, prompt injection, etc.
AI-specific threat response framework

Source: Financial Services Commission press release (2025.12)

FSS AI Risk Management Framework (AI RMF)

The Financial Supervisory Service introduced the **"Financial AI Risk Management Framework (AI RMF)"** in January 2026. Currently, **118 financial institutions operate 653 AI services**, but approximately **85% lack AI ethics principles and risk management standards**.

AI RMF — 3 Domains

1. Governance	2. Risk Assessment	3. Risk Control
Establish AI decision-making body and dedicated organization	Risk identification/measurement -> mitigation -> residual risk	Pre-launch risk mitigation verification
Develop risk management regulations	**High/Medium/Low** 3-tier classification	Operational-phase monitoring standards
Independently separated from planning/development	Quantitative assessment of legality, reliability, good faith, security	Re-classification on risk changes

Comparison with EU AI Act

Category	Korea AI Basic Act	EU AI Act
Effective Date	2026. 1. 22.	High-risk AI: 2027. 12.
Classification	General/High-impact (2 tiers)	Prohibited/High-risk/Limited/Minimal (4 tiers)
Finance Application	Credit scoring/loan assessment explicit	Credit scoring/insurance pricing included
Prohibited AI	None	Social scoring, real-time remote biometrics, etc.
Obligated Parties	Single operator-centric standard	Entire supply chain: providers, deployers, users
Penalties	Fine up to 30M KRW	Up to 7% global revenue or 35M EUR
Approach	Promotion-oriented + self-regulation	Safety-oriented + mandatory compliance
Impact Assessment	Best-effort obligation	Legally mandatory pre-conformity assessment

Comparison Implications

Korea's AI Basic Act has penalties **thousands of times lower** than the EU and relies more on self-regulation, but this is a **soft-landing strategy for the initial implementation period**. The FSC Integrated Guidelines have a broader scope than the AI Basic Act, requiring **prior notification for all customer-facing AI services**. Regulatory intensity is expected to gradually increase through future Enforcement Decree amendments and FSC detailed regulations, making **proactive preparation** advantageous.

Global Regulatory Trends Summary

Global regulation of financial AI is converging in three directions.

Transparency & Explainability

Expanding AI disclosure obligations
Requiring decision rationale explanations
Guaranteeing right to refuse automated decisions

Fairness & Non-discrimination

Quantitative bias measurement by protected attributes
Market herding and conflict of interest prevention
Mandating periodic fairness audits

Safety & Accountability

Risk-based classification (high-risk AI)
Mandatory emergency shutdown mechanisms
Audit trail retention obligations

FSS Guidelines Mapping — 7 Principles and System Response

Mapping the current response status of the PLE-based financial AI recommendation system against the FSC Integrated AI Guidelines 7 Principles and FSS AI RMF requirements.

Per-Principle Response Matrix

7 Principles	System Response	Level
1. Governance (AI RMF G-1~G-6)	3-tier governance framework design complete (Decision Committee -> Operations Team -> Internal Audit) Monthly/quarterly governance report auto-generation 36-item regulatory compliance registry auto-check	◯
2. Legality (AI RMF R-2)	Eligibility/suitability auto-verification (Financial Consumer Protection Act Art. 17/18) PII de-identification (SHA-256 + domain-specific salt) Credit Information Act retention compliance (5 years)	●
3. Auxiliary Nature (AI RMF C-4)	Human oversight + Kill Switch 3-tier emergency shutdown Human reprocessing routing (P1/P2/P3 SLA) Right to refuse AI automated decisions + alternative pathway	●
4. Reliability (AI RMF R-3, C-2~C-3)	Champion-Challenger automatic model competition PSI-based drift detection + 3-consecutive-day retraining trigger IG-based per-feature attribution + natural language recommendation rationale	●
5. Financial Stability (AI RMF R-3~R-4)	HHI/Gini/Entropy herding detection DI/SPD/EOD fairness 3-metric auto-measurement 5 protected attributes (age, gender, region, income, lifecycle)	●
6. Good Faith (AI RMF R-4)	AI disclosure centralized management + per-segment notification separation Conflict of interest prevention (auto-penalty when high-fee products exceed 40%) Opt-out registration/withdrawal/confirmation full lifecycle management	●
7. Security (AI RMF R-5)	Prompt injection defense 8 patterns (4 Korean + 4 English) Model integrity SHA-256 hash verification HMAC + hash chain audit log immutability	●

● Met ◯ Partially met (organizational decision needed)

AI RMF Domain-Level Detailed Mapping

Domain 1: Governance (G-1 G-6)

No.	RMF Requirement	Current Response Status	Level
G-1	Establish AI top decision-making body	3-tier governance framework designed; formal committee establishment needed	△
G-2	Independent AI risk management org	Operations team handles both dev and ops; independent function separation needed	△
G-3	Establish AI risk management regulations	FD-TVS, drift detection logic exists; internal regulation documentation needed	◯
G-4	Establish and publish AI ethics principles	Not yet established — enterprise AI ethics document needed	△
G-5	High-impact AI pre-approval process	Pre-launch approval process formalization needed	△
G-6	Periodic AI utilization reporting	Monthly report auto-generation complete; reporting line formalization needed	◯

Domain 2: Risk Assessment (R-1 R-6)

No.	RMF Requirement	Current Response Status	Level
R-1	Risk classification per AI service	Self-assessment of high-impact AI applicability complete; formal classification system needed	◯
R-2	Legality assessment	Eligibility auto-verification + de-identification + retention compliance	●
R-3	Reliability assessment	Champion-Challenger + drift detection + fairness auto-measurement	●
R-4	Good faith assessment	5 protected attributes DI/SPD/EOD + conflict prevention + Parquet archiving	●
R-5	Security assessment	Local AI + SHA-256 encryption + prompt injection defense + integrity verification	●
R-6	Residual risk assessment	FD-TVS risk penalty auto-block; Risk Appetite documentation needed	◯

Domain 3: Risk Control (C-1 C-6)

No.	RMF Requirement	Current Response Status	Level
C-1	Pre-launch risk mitigation verification	Eligibility rules + fatigue filtering + A/B testing framework	●
C-2	Operational monitoring standards	Drift auto-detection + performance dashboard + anomaly alerts	●

C-3	Periodic model evaluation/retraining	Weekly/monthly auto-retraining + Champion-Challenger + MLflow versioning	●
C-4	Human oversight framework	Kill Switch 3-tier + human reprocessing routing + opt-out	●
C-5	Emergency shutdown mechanism	GLOBAL/PER_TASK/PER_CLUSTER 3-tier kill switch	●
C-6	Audit trail assurance	HMAC hash chain + S3 Object Lock + 7 audit tables	●

EU AI Act Article-Level Mapping

The EU AI Act is likely to classify financial AI recommendation systems as **high-risk AI** (Annex III, 5(b): credit scoring/insurance pricing). The following maps system responses to key articles.

Article 13: Transparency and Information Provision

Art. 13 Requirements and System Response

Art. 13 Requirement	System Response
Notify use of high-risk AI	AI disclosure centralized management + per-segment notification separation All recommendation outputs automatically include AI usage labeling
Explain system operation	2-Layer recommendation rationale (L1 Template + L2 LLM) 3-Agent pipeline (Feature Selector → Reason Generator → Safety Gate)
Disclose input data specification	Feature schema auto-documentation (316D current / 734D full-bank design) Training data source, scope, and pseudonymization status recorded
Disclose performance level	Model card auto-generation (architecture, performance, feature importance) Per-task independent AUC tracking (14 tasks)
Log generation obligation	HMAC + hash chain audit log (S3 Object Lock WORM) Full recommendation history Parquet archiving

Article 14: Human Oversight

Art. 14 Requirements and System Response

Art. 14 Requirement	System Response
Provide interface for humans to oversee AI system	Offline: Staff screen shows recommendation list + rationale; final recommendation is staff judgment Online: AI operations team directly controls recommendation direction, targets, and exclusion lists
Ensure ability to intervene and halt system	Kill Switch 3-tier (GLOBAL/PER_TASK/PER_CLUSTER) Auto-switch to conservative mode on drift detection

Right to refuse automated decisions + alternative pathway	Opt-out registration/withdrawal/confirmation + 3-tier human reprocessing routing (P1 urgent 1h / P2 4h / P3 24h SLA)

Article 15: Accuracy, Robustness, and Cybersecurity

Art. 15 Requirements and System Response

Art. 15 Requirement	System Response
Achieve and maintain appropriate accuracy levels	Champion-Challenger automatic model competition PSI-based drift detection + auto-retraining on 3 consecutive critical days
Robustness against errors and faults	Ablation-based per-component contribution quantification Distilled LGBM fallback (auto-switch on teacher performance anomaly)
Cybersecurity threat response	Prompt injection defense 8 patterns (5 severity levels) Model integrity SHA-256 verification Input validation (format + size limits)
Bias prevention measures	DI/SPD/EOD 3-metric fairness auto-measurement 5 protected attributes continuous monitoring Auto-incident creation on threshold violation

GDPR Article 22: Automated Decision-Making

GDPR Art. 22 and Related Provisions Response

GDPR Art. 22 grants data subjects the right to refuse **"decisions based solely on automated processing, including profiling."** Korea's Personal Information Protection Act Art. 37-2 guarantees the same right.

GDPR Requirement	System Response
Right to refuse automated decisions (Art. 22(1))	Opt-out registration/withdrawal/confirmation full lifecycle management Immediate switch to human alternative pathway upon refusal
Right to request human intervention (Art. 22(3))	Human reprocessing routing (auto-counselor switch for 7 reason categories) P1/P2/P3 SLA management
Profiling information provision (Art. 13(2)(f))	Decision criteria auto-documentation (30+ filter reasons with explanations) Feature reverse mapping + per-task interpretation

Right to erasure (Art. 17)	30-day PII retention policy + encrypted deletion S3 Lifecycle auto-applied
Data subject rights audit (Art. 30)	Among 7 audit tables: profiling_audit, opt_out_audit, consent_audit DynamoDB serverless management + TTL auto-cleanup

Korea AI Basic Act High-Impact AI Classification and Financial Application

High-Impact AI Applicability Assessment

Under current law, financial product recommendation systems are **not subject to explicit high-impact AI classification as they do not constitute loan assessment**. However, expanded designation is possible under the following scenarios.

Scenario	Likelihood	Basis
Enforcement Decree amendment to include all financial product recommendations	Medium	EU AI Act already covers this scope
Customer classification/segmentation interpreted as rights evaluation	Low-Medium	Similar structure to investment suitability assessment
Business expansion to insurance product recommendations	High	Insurance pricing is EU high-risk and domestic high-impact in both frameworks
Discriminatory benefits based on churn prediction	Medium	Directly linked to fairness issues
FSC guidelines separately regulating recommendation AI	Medium-High	Must monitor trends after integrated guidelines take effect

> **Response Strategy**
>
> While not currently subject to direct high-impact AI classification, considering the **direction of the regulatory environment**, proactively building a governance framework at **virtually the same level as high-impact AI** is a rational strategy. This not only minimizes regulatory risk but also provides an immediately deployable foundation for future business expansion (insurance, fund recommendations, etc.).

AI Basic Act Key Article Response Status

Requirement	Basis	Current Response	Level
Prior notification of AI use	Art. 31	AI disclosure centralized management, per-segment notification separation Automatically included in all recommendation outputs	●

Labeling of AI-generated content	Art. 31	AI generation label auto-applied to recommendation rationale text	●
Risk management measures implementation	Art. 32	FD-TVS + Kill Switch (3-tier) + eligibility/ suitability verification 1. drift detection	●
High-impact AI applicability check	Art. 33	Scenario-based applicability analysis complete	●
Risk management and user protection	Art. 34	Safety/trust document + model card auto-generation 3-stage recommendation rationale pipeline	●
Impact assessment execution	Art. 35	System documentation complete; periodic execution process needs organizational establishment	◯
Right to refuse automated decisions	PIPA Art. 37-2	Opt-out + 3-tier human reprocessing routing	●
Governance framework establishment	7 Principles	3-tier framework designed; formal committee establishment needed	◯

● Met ◯ Partially met △ Insufficient

Regulatory Timeline

Key Regulatory Dates

Date	Event
2025. 1. 21	AI Basic Act promulgation
2025. 08	AI Basic Act Enforcement Decree pre-announcement
2025. 12. 22	FSC Integrated AI Guidelines (draft) release
2026. 1. 15	FSS AI RMF introduction
2026. 01. 22	**AI Basic Act effective**
2026 Q1	Integrated Guidelines and AI RMF finalized and effective
2027. 01 (est.)	AI Basic Act fines enforcement begins
2027. 12	EU AI Act high-risk AI provisions fully applicable

Compliance Architecture

Overall Structure

The regulatory compliance infrastructure is organized in 3 layers.

> **Layer 1: AWS Native (Automatic, No Additional Cost)**
>
> - **CloudTrail** -> All AWS API calls automatically recorded
> - **S3 Versioning** -> Data/model change history preserved
> - **S3 Object Lock** -> Audit log immutability (WORM)
> - **KMS** -> Encryption key management + key usage audit
> - **IAM Access Log** -> Resource access audit

> **Layer 2: Platform Level (Semi-automatic)**
>
> - **ComplianceAuditStore** -> 7 audit tables (DynamoDB)
> - **AuditLogger** -> HMAC + hash chain (S3 + DynamoDB)
> - **DataLineageTracker** -> Feature-to-source tracing (S3 metadata)
> - **ExperimentTracker** -> Experiment metrics (SageMaker Experiments)
> - **DriftDetector** -> PSI monitoring (SageMaker Model Monitor)

> **Layer 3: Business/Regulatory Level (Explicit)**
>
> - **RegulatoryComplianceChecker** -> 36-item auto-check
> - **FairnessMonitor** -> DI/SPD/EOD protected attribute monitoring
> - **HerdingDetector** -> Systemic risk herding detection
> - **IncidentReporter** -> Auto-report by severity
> - **GovernanceReportGenerator** -> Monthly/quarterly governance reports
> - **KillSwitch** -> Emergency model deactivation

Audit Trail

Immutable Logs — HMAC + Hash Chain

Each audit log entry is signed with HMAC-SHA256, and consecutive entries are linked via SHA256 hash chain (prev_hash) to prevent tampering.

On-Prem	AWS
Local JSONL + HMAC + hash chain	S3 Object Lock (WORM) + HMAC + hash chain

AWS Reinforcement:

- S3 Object Lock: Physical deletion blocked at AWS level (even administrators cannot delete)
- KMS encryption: Audit logs themselves are encrypted
- CloudTrail: Access to audit logs is also audited (meta-audit)

7 Regulatory Audit Tables

DynamoDB-based serverless management (auto-scaling, per-item TTL):

Table	Purpose	Retention
ks_audit	Kill switch activation/deactivation history	7 years
consent_audit	Marketing consent change history (grant/revoke/renew)	7 years
profiling_audit	Data subject rights exercise (access/correction/deletion/restriction/portability)	7 years
opt_out_audit	AI decision refusal history	7 years
incident_audit	Regulatory incidents (by severity)	7 years
distillation_audit	Teacher-student model performance gap	3 years
embedding_audit	Embedding quality metrics	3 years

Fairness Monitoring

3 fairness metrics are continuously monitored across 5 protected attributes.

Protected Attributes

- **Age group**: youth / middle / pre_senior / senior
- **Gender**: M / F / unspecified
- **Region type**: metropolitan / urban / rural
- **Income quintile**: low / middle / high
- **Lifecycle**: 6 classes

Thresholds and Actions

- **DI (Disparate Impact)**: 0.8 1.25
- **SPD (Statistical Parity Difference)**: |SPD| <= 0.1
- **EOD (Equal Opportunity Difference)**: |EOD| <= 0.1

DI < 0.6 -> CRITICAL incident
DI < 0.8 -> MAJOR incident
|SPD| > 0.1 -> MINOR incident

Drift Surveillance

Combines SageMaker Model Monitor (default) + custom PSI (extended).

Metric	Threshold	Auto Action
PSI (warning)	0.1	Warning logging
PSI (critical)	0.25	Alert responsible party

3 consecutive critical days	–	Auto-retraining trigger (Step Functions)

Per-feature individual PSI is tracked, enabling early detection of sudden changes in specific features even when the overall distribution appears stable.

Herding Detection

Prevents **systemic risk** from excessive recommendation concentration on the same products.

Metric	Description	Severity Classification
HHI	Herfindahl-Hirschman Index — market concentration	none -> low -> medium
Gini coefficient	Recommendation inequality measurement	-> high -> critical
Entropy	Recommendation diversity measurement	
Herding rate	Proportion of most-recommended product	Kill switch review on critical

Kill Switch

3-tier emergency model deactivation system.

Kill Switch Structure

GLOBAL
Full model deactivation
State stored in DynamoDB
Checked on every request

PER_TASK
Specific task only deactivated (e.g., click, purchase)
Only that task falls back

PER_CLUSTER
Specific customer segment only (e.g., cluster_5)
Only that cluster falls back

Fallback strategies: Rule-based recommendations / previous model rollback / recommendation deactivation (choose one)

Opt-out and Consent Management

Feature	Description
AI decision refusal registration	Opt-out registration -> immediate switch to human alternative pathway
Refusal withdrawal	Previous opt-out withdrawn -> AI recommendation reactivated
Marketing consent management	Grant/revoke/renew history tracking, per-channel and per-purpose opt-in verification
Nighttime SMS blocking	Auto-filtering by marketing consent status + time zone

Audit trail	All consent changes and refusal history recorded in consent_audit, opt_out_audit

Data Retention Policy

Automatically applied via S3 Lifecycle Rules.

Category	Retention	Action	Regulatory Basis
Raw Data	30 days	Delete	GDPR minimization principle
Processed Features	90 days	Glacier archive	Retraining support
Training Data	365 days	Glacier archive	Reproducibility assurance
Model Checkpoints	365 days	Retain	Rollback support
Inference Results	90 days	Delete	Dispute response
Audit Logs	2,555 days (7 years)	Immutable retention (WORM)	Financial regulation 7-year retention
PII Data	30 days	Encrypted deletion	GDPR Art. 17 right to erasure

Incident Management

Automated classification and response system by severity.

Severity	Response Time	Trigger	Report To
CRITICAL	1 hour	Kill switch activation, DI<0.6, security breach	MSIT/FSS/CISO
MAJOR	4 hours	DI<0.8, herding critical, model rollback	FSS/AI Committee
MINOR	24 hours	Drift warning, quality degradation, herding high	ML Team

Bedrock Data Protection Architecture

The most critical regulatory issue when leveraging LLMs in financial AI systems is the prevention of customer data leakage. Amazon Bedrock addresses this through five structural safeguards:

Safeguard	Details
No Data Training	Input/output data is never transmitted to model providers (Anthropic, Upstage, Meta, etc.) and is never used for model retraining (including fine-tuning). AWS guarantees this contractually in its Terms of Service.
Transit Encryption	TLS 1.2+ encryption protects data in transit.

VPC PrivateLink	Bedrock API is invoked through VPC-internal endpoints without traversing the public internet. Customer data is never exposed to public networks.
In-Region Processing	All inference is processed in the ap-northeast-2 (Seoul) Region. Customer data never leaves Korea.
CloudTrail Audit	Every Bedrock API call (InvokeModel, Converse, etc.) is automatically recorded in CloudTrail. Complete audit trail of who called which model when.

Regulatory Mapping

Regulation	Requirement	Bedrock Compliance
PIPA (Korea)	Third-party provision vs. delegated processing	Bedrock constitutes delegated processing within AWS infrastructure. Data is not transmitted to model providers, thus not third-party provision.
PIPA (Korea)	Cross-border transfer restriction	In-region processing in ap-northeast-2. No cross-border transfer occurs.
Korean FSS AI Guidelines	Data governance	CloudTrail audit logs + VPC isolation + transit encryption enable complete data flow tracking.
EU AI Act Art.10	Data governance	No-training-use guarantee. Inference data processing location documented.
AI Basic Act (Korea)	High-impact AI data management	HMAC audit logs and CloudTrail dual-recording prove data processing history.

Data Flow Diagram

Data flow during recommendation reason generation and agent diagnostics:

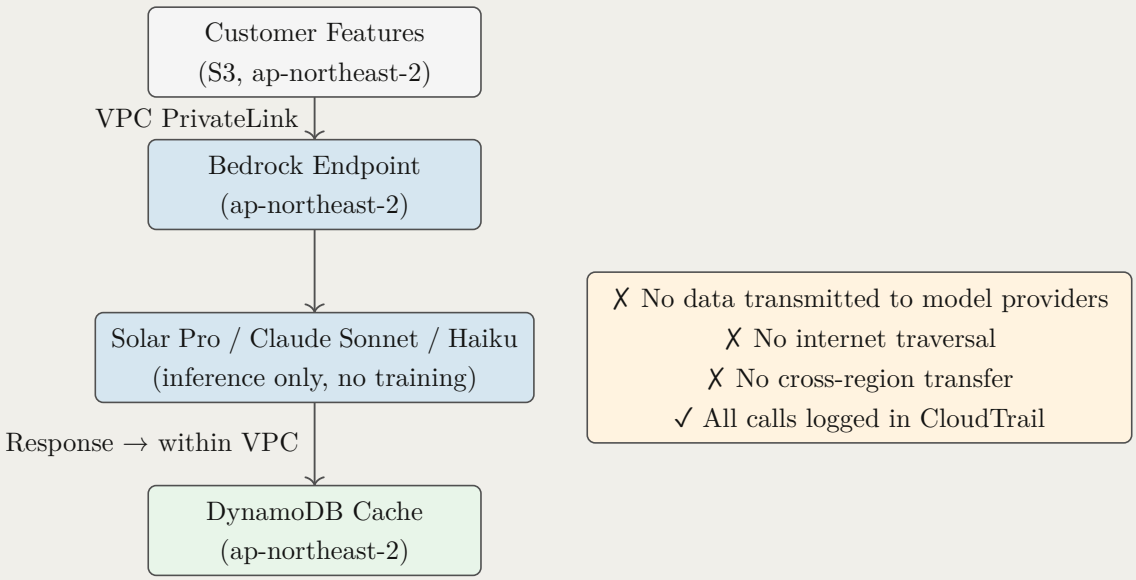

Figure 1: AWS Bedrock data flow: processed via VPC PrivateLink with no cross-region data transfer.

On-Premises Regulatory Compliance

Air-gapped on-premises environments are **structurally stronger** than cloud deployments from a data protection perspective, as external network access is architecturally impossible.

Data Protection

Item	On-Premises	AWS (Bedrock)
External data transfer	**Impossible** — air-gapped	In-region via VPC PrivateLink
Model provider data access	**N/A** — local open-source models	Contractually prohibited by Bedrock ToS
Audit trail	Local HMAC hash-chain (JSONL)	S3 Object Lock + CloudTrail
Cross-border transfer	**Cannot occur**	Processed within ap-northeast-2

Model Configuration

All on-premises models are open-source:

Purpose	Model	License & Notes
Reason generation/critique	Exaone 3.5 7.8B	Apache 2.0 (LG AI Research). Korean-specialized.
Agent consensus	Qwen 2.5 14B Q4	Apache 2.0 (Alibaba). Logical reasoning.
Embeddings	all-MiniLM-L6-v2	Apache 2.0 (sentence-transformers).

Sequential loading on RTX 4070 (12GB VRAM). Zero vendor lock-in; model replacement requires only config changes.

Ops/Audit Agents

On-premises agents use the **same rule engine, checklist, and tool catalog** as AWS. Differences:

Feature	On-Premises	AWS
Checklist evaluation	Identical (48 items)	Identical
Consensus mechanism	2-Round hybrid (Qwen 14B × 5+2)	Independent voting (Sonnet × 3)
Minority report	Identical — locked at Round 1	Identical
Operator dialog	Not available — structured reports only	Sonnet Tool Use conversation
Case store	Identical (LanceDB)	Identical
Notifications	Email/Slack	SNS + Slack

Regulatory Compliance Comparison

Requirement	On-Premises	AWS
PIPA cross-border transfer	Impossible (air-gapped)	In-region processing
FSS data governance	Local HMAC audit logs	CloudTrail + S3 Object Lock

EU AI Act human oversight	Agent recommendation + human decision (same)	Same + dialog interface
AI Basic Act kill switch	Local kill switch (same)	DynamoDB kill switch
Explainability	IG-based reasons + Exaone rewrite	IG-based reasons + Solar rewrite

On-premises lacks conversational agent capabilities but offers structurally perfect data protection. From a regulatory perspective, "customer data never leaves the premises" is the strongest possible safeguard.

Model Risk Management (MRM) Framework

Full-lifecycle model governance aligned with **SR 11-7** (Federal Reserve/OCC), **EBA ML Guidelines**, and **NIST AI RMF 1.0**.

MRM Lifecycle

Development → Validation → Approval → Monitoring → Retrain / Retire

Stage	Activity	System Component
Development	Feature engineering, model training, offline evaluation	PipelineRunner (Phase 0) + train.py (Phase 1)
Validation	Champion-Challenger comparison, ablation, fairness audit	ModelCompetitionManager + FairnessMonitor
Approval	Manual review gate — `auto_promote = false`	AI Committee sign-off (Human-in-the-Loop)
Monitoring	Drift, performance, fairness, herding — continuous	DriftDetector + PerformanceMonitor + FairnessMonitor
Retrain / Retire	Triggered retrain or model decommission	ConsecutiveDriftTracker → dag_monthly_retrain

Model Inventory

Model Registry

Every model version is recorded in `ModelCompetitionManager.model_registry` with:

- **Model ID, version, training date, dataset snapshot hash**
- **Architecture config** (expert count, task groups, adaTT strengths)
- **Validation metrics** (per-task AUC/F1/MAPE, calibration, fairness DI/SPD/EOD)
- **Approval status** (pending / approved / rejected / retired)
- **Lineage** — which champion it replaced and why

Independent Validation (Champion-Challenger)

SR 11-7 Pillar 2 — Independent Model Validation

Gate policy: `auto_promote = false` — no model enters production without explicit human approval.

Comparison criteria:

1. Per-task AUC / F1 / MAPE (absolute performance)
2. Paired t-test across temporal folds (statistical significance)
3. Calibration: predicted probability vs. observed frequency
4. Fairness: Disparate Impact (DI > 0.8), SPD, EOD

> 5. Inference latency and cost efficiency
>
> **Decision flow**: Challenger trained → metrics collected → report generated → AI Committee reviews → promote / reject / request re-experiment.

Continuous Monitoring → Retrain Trigger

Monitor	Trigger Condition	Action
DriftDetector (PSI)	Feature or prediction PSI > 0.25	Alert (Orange)
ConsecutiveDriftTracker	3 consecutive days of critical drift	Auto-trigger `dag_monthly_retrain`
FairnessMonitor	DI < 0.8 or SPD/EOD threshold breach	Alert → AI Committee review
PerformanceMonitor	Task AUC below Yellow threshold	Alert → retrain evaluation

After retraining completes, `dag_champion_challenger` is invoked automatically: the new model becomes the challenger and must pass the validation gate described above before promotion.

Emergency Response (Kill Switch)

When monitoring detects a critical failure, the 3-tier Kill Switch provides immediate response:

1. **Deactivate** — model serving is halted (global / per-task / per-cluster granularity)
2. **Rollback** — previous approved model version is restored from the registry
3. **Fallback** — rule-based recommendation engine takes over until a validated model is re-deployed

All kill-switch activations are logged as CRITICAL incidents and reported to MSIT/FSS/CISO within 1 hour per the Incident Management protocol above.

Operational and Audit Agents

Architecture Overview

The serving agents (L1 Rule / L2a Retrieval / L2b Generation) reside on the real-time recommendation path. The **OpsAgent** and **AuditAgent** described here are fully separate **batch-only** agents that share no state with the serving path. They execute exclusively as downstream tasks within Airflow DAGs. This separation ensures regulatory surveillance functions do not impact real-time serving SLAs.

OpsAgent (Operational Agent)

Trigger and Input

Trigger: `dag_drift_monitoring` completion, training Job completion event
Input data:
- `eval_metrics.json`: per-task AUC/F1/MAPE, calibration metrics
- Training logs: per-task loss trajectory, gradient norm history
- Gate entropy: expert routing skew (MoE gate softmax distribution)
- PSI drift reports: feature/prediction/label distribution changes

Processing pipeline: JSON parsing → anomaly detection rules (threshold-based pre-filtering) → on anomaly detection, forward structured prompt to LLM.

Prompt example

"Analyze the following eval_metrics.json and gate entropy. Report (1) tasks with degraded performance, (2) gate skew (experts with entropy < 0.3), (3) features exceeding PSI threshold, (4) recommended actions in structured format."

Output: Model health report (Markdown) — stored at S3 `governance/ops_reports/`. On anomaly detection, alerts are sent via Slack channel and email.

AuditAgent (Audit Agent)

Trigger and Input

Trigger: `dag_fairness_monitoring` completion, quarterly governance cycle (`dag_governance_quarterly`)
Input data:
- FairnessMonitor report: DI/SPD/EOD (5 protected attributes × per-task)
- Audit trail integrity verification: HMAC hash chain validation status
- Opt-out statistics: request count, processing rate, avg. processing time by period
- Governance checklist: 36-item auto-check results

Processing pipeline: automated comparison against regulatory thresholds (DI ≥ 0.8, $|\text{SPD}| \leq 0.1$, $|\text{EOD}| \leq 0.1$) → violation extraction → request LLM to summarize violations and recommend actions.

> **Prompt example**
>
> "Compare the following fairness metrics against FSS thresholds (DI$\geq$ 0.8, |SPD|$\leq$ 0.1, |EOD|$\leq$ 0.1). Report (1) violations (protected attribute, task, metric value), (2) severity (P1/P2/P3), (3) recommended actions."

Output: Regulatory compliance report (Markdown) — stored at S3 `governance/audit_reports/`. Alerts dispatched by priority: P1 (immediate escalation), P2 (review within 24h), P3 (included in quarterly report).

Model Selection

Environment	Model	Rationale
Air-gapped (on-prem)	Exaone 3.5 7.8B (reason gen.) + Qwen 2.5 14B Q4 (agent consensus)	Apache 2.0 open-source. Sequential loading on RTX 4070.
Cloud (AWS)	Claude Haiku 4.5 API	Cost-efficient ($0.25/1M input), reliable structured output

Cost estimate: 1–2 calls per batch yields approximately $0.01/day. Input consists of JSON metrics (a few KB) + prompt, so token consumption is minimal.

Design Principles

Automation scope

Batch-only: does not intervene in real-time serving path
Structured judgments: threshold comparison, trend analysis, violation classification are handled by the agent
Unstructured judgments: suspected data contamination, business context shifts, regulatory interpretation are reserved for humans

Audit artifact guarantees

HMAC signing: agent output itself is an audit artifact — HMAC-signed and immutably stored upon generation
Hash chain: tamper-proof guarantee via hash chain linked to prior reports
Human review: "check when you arrive" model — agent organizes, humans review at their own pace

Airflow DAG Integration

DAG	Agent task	Cadence
`dag_drift_monitoring`	After drift detection → `ops_agent_report` task appended	Daily
`dag_fairness_monitoring`	After fairness measurement → `audit_agent_report` task appended	Daily

`dag_governance_quarterly`	After 36-item check → `audit_agent_quarterly` report	Quarterly
Training Job completion callback	`ops_agent_training_report` — training metric analysis	Event-driven

Human-in-the-Loop Design

The FSS ultimately requires **human-involved decision-making**. EU AI Act Art. 14 also mandates the same principle. Even when everything is automated, final review is performed by humans.

Design Principle

> **AI is the tool; final decision authority rests with humans**
>
> The PLE-based recommendation system is not "replacing humans" but a tool that "supports human decision-making." This principle is embedded throughout the system architecture.

Per-Channel Human Oversight Framework

Channel	Human Oversight Method
Offline (Branch)	AI **provides** recommendation list + rationale on staff screen Final recommendation decision is made by **staff judgment** Staff reviews rationale and recommends based on customer situation
Online (App/Web)	AI auto-serves recommendation list AI operations team **directly controls** recommendation direction, targets, and exclusion lists Immediate shutdown via kill switch in emergencies

5 Human Intervention Points

1. Recommendation Rationale Sampling Review

While exhaustive review is infeasible, recommendation rationale outputs are periodically sampled and reviewed for quality, suitability, and regulatory compliance. The recommendation quality monitoring system (L1 rules / L2a rewrite / L2b LLM verification) generates automatic quality metrics, triggering human review on anomaly detection.

2. Model Replacement Approval

Champion-Challenger comparison results are confirmed and approved by humans. Even for auto-replacement, a report including replacement rationale, performance comparison, and fairness metrics is generated, with post-review by the operations team.

3. Incident Escalation

Auto-detected anomalies (fairness violations, drift, herding) are assessed and acted upon by humans. CRITICAL incidents trigger automatic kill switch activation, followed by human root cause analysis and recovery.

4. Fairness Review

Fairness monitoring results are periodically reviewed by humans. Domain experts provide context-aware interpretation of DI/SPD/EOD auto-measurement results, and decide on threshold adjustments or model retraining as needed.

5. Opt-out and Complaint Handling

Human reprocessing pathways are automatically activated for customer AI decision refusals and complaints. Auto-counselor routing occurs for 7 reason categories, managed under P1(1h)/P2(4h)/P3(24h) SLA.

Human Reprocessing Reason Classification

P1 (Urgent 1 hour)

- Compliance violation detected
- Customer failing suitability check
- Kill switch activation situation

P2 (4 hours) / P3 (24 hours)

- AI decision refusal (opt-out)
- Low-confidence recommendation results
- Explanation supplementation request
- General complaints

Automated Governance Report Generation

Report Framework

Auto-generated on a monthly/quarterly basis, stored in S3, and distributed to the AI Governance Committee.

9-Section Structure

Section	Content	Data Source
1	Fairness summary (DI/SPD/EOD by protected attribute)	FairnessMonitor
2	Drift summary (PSI statistics for the period)	DriftDetector
3	Incident summary (CRITICAL/MAJOR/MINOR counts + details)	IncidentReporter
4	Model change history (training/deployment/rollback)	ExperimentTracker
5	Kill switch history (activation/deactivation counts)	ComplianceAuditStore
6	Recommendation quality (L1/L2 rationale quality metrics)	ReasonQualityMonitor
7	Risk trends (herding trends, risk levels)	HerdingDetector
8	Audit store summary (7 table record counts)	ComplianceAuditStore
9	Executive summary (auto-generated narrative)	Combined

36-Item Regulatory Compliance Auto-Check

A quarterly full check is automatically executed via the 36-item regulatory compliance registry.

Group A (18 Items Implemented)

Model card, training data documentation, safety/trust document
AI disclosure, explainability, opt-out rights
Human review, kill switch, model rollback
Health check, fairness monitoring, suitability constraints, etc.

GAP Group (18 Items Gap Analysis)

Drift auto-response, recommendation rationale quality verification
SLA tracking, EU AI Act high-risk classification
PIA (Privacy Impact Assessment), etc.

Check frequency:

- Daily: Kill switch status, health check
- Weekly: Fairness metrics, explainability

- Quarterly: Full 36-item comprehensive check

3-Tier Governance Framework

Tier 1: Decision — AI Risk Management Committee

AI utilization policy approval, high-impact AI applicability determination, annual impact assessment approval, critical incident reporting framework

Tier 2: Execution — AI Operations Team

Model development and operations, performance monitoring, fairness review, recommendation quality management, incident response

Tier 3: Verification — Internal Audit & Compliance

Independent audit, regulatory compliance verification, impact assessment review, external audit response

Comprehensive Regulatory Mapping

Regulation	Relevant Articles	Design Reflection
AI Basic Act	Art. 31 (AI content labeling)	AI disclosure + recommendation rationale L1/L2
	Art. 33 (high-impact AI governance)	36-item registry + governance report
	Art. 34 (risk management records)	Audit log immutability + 7 audit tables
Financial Consumer Protection Act	Art. 19 (explanation duty)	Feature reverse mapping + per-task interpretation
FSS AI RMF	(1) Legality	36-item auto-check
	(2) Safety/Trust	Kill switch + incident reporting
	(4) Reliability	Drift surveillance + auto-retraining
	(5) Financial Stability	Fairness DI/SPD/EOD + herding detection
GDPR	Art. 17 (right to erasure)	30-day PII retention + encrypted deletion
	Art. 22 (automated refusal)	opt_out_audit table
	Art. 35 (DPIA)	PIA gap analysis items
PIPA	Art. 28-2 (pseudonymized data)	Pseudonymization processing record audit

	Art. 37-2 (automated decisions)	Opt-out + human reprocessing
EU AI Act	Art. 13 (transparency)	Recommendation rationale + model card + audit log
	Art. 14 (human oversight)	Kill switch + human reprocessing + per-channel oversight
	Art. 15 (accuracy/security)	Drift detection + prompt injection defense

This document was prepared as of April 2026 and may be updated in accordance with the finalization of the AI Basic Act Enforcement Decree and FSC Integrated AI Guidelines. It is recommended to update this document and report to the AI Risk Management Committee when regulatory changes occur.

Ops/Audit Agent Integration

All regulatory compliance components documented herein (FairnessMonitor, HerdingDetector, ComplianceChecker, AuditLogger, etc.) are wrapped as tools in the AuditAgent's 48-item checklist for automated inspection.

A 3-agent consensus mechanism (Sonnet×3 independent voting) structurally mitigates hallucination, and minority reports preserve dissenting opinions. Diagnostic history accumulates in a LanceDB case store, serving as "continuous improvement evidence" for regulatory audits.

Core design principle: **"AI analyzes, humans decide"** — agents recommend only; final decisions are made by operators. This structurally satisfies EU AI Act Art.14 (human oversight), Korean FSS AI guidelines (human intervention), and AI Basic Act (kill switch).

Detailed design: Design Document 11 (`docs/design/11_ops_audit_agent.typ`)

Temporal Fact Store (Added 2026-04)

A core audit evidence requirement is **"reproducing system state at a specific point in time"**. Questions like "At 2026-03-15, when customer A received a recommendation, what were the model version, features, and verdicts?" traditionally require joining across distributed components (AuditLogger, DiagnosticCaseStore, `pipeline_state.json`) — expensive.

We solved this with `TemporalFactStore`, adapted from the Zep/Graphiti pattern. The schema is `(entity, attribute, value, valid_from, valid_to)`, and most audit queries are **single-entity point-in-time reconstructions** resolvable via native LanceDB filters.

Property	Implementation
No deletion	`expire_fact()` sets `valid_to` only, originals preserved
Backend	Shares LanceDB instance with `DiagnosticCaseStore` (zero new deps)
Z-suffix compat	Normalizes diverse timestamp formats from external callers

Query API	`snapshot_at()`, `get_timeline()`, `expire_fact()`

This satisfies EU AI Act Art.12 (record-keeping), Korean FSS AI guidelines (audit trail), and AI Basic Act temporal evidence requirements via a single store.